# MILITARY SPECIFICATION

## INDICATOR, ATTITUDE ARU-18/A, DIRECTOR

### 1. SCOPE

1.1 This specification covers one type of remote attitude director indicator, designated ARU-18/A.

### 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| PPP-B-636 | Box, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-C-675 | Coating of Glass Optical Elements (Anti-Reflection) |
| MIL-E-5400 | Electronic Equipment, Aircraft, General Specification for |
| MIL-C-5541 | Chemical Films for Aluminum and Aluminum Alloys |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-S-6872 | Soldering Process, General Specification for |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification for |
| MIL-C-7769 | Cushioning Material, Bound Fiber |
| MIL-A-8625 | Anodic Coatings, for Aluminum and Aluminum Alloys |
| MIL-L-25467 | Lighting, Integral, Instrument, General Specification for |
| MIL-M-26512 | Maintainability Program Requirements for Aerospace Systems and Equipment |
| MIL-R-26667 | Reliability and Longevity Requirements, Electronic Equipment, General Specification for |
| MIL-L-27160 | Lighting, Instrument, Integral, White, General Specification for |
| MIL-D-70327 | Drawings, Engineering and Associated Lists |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of U. S. Military Property |



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|---|---|
| MIL-STD-143 | Specifications and Standards, Order of Precedence for the Selection of |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-810 | Environmental Test Methods for Aerospace and Ground Equipment |
| MIL-STD-838 | Lubrication of Military Equipment |
| MS17322 | Meter, Time Totalizing Miniature, Digital 115 Volt 400 Cycle |
| MS24367 | Lamp – Incandescent, Miniature, Integral Lighting |
| MS24515 | Lamp – Sub Miniature |
| MS25237 | Lamp, Incandescent, T-1-3/4 Bulb, Midget Flange Base |
| MS33545 | Case – Instrument 5 x 5-1/5 Standard Dimensions for |
| MS33558 | Numerals and Letters, Aircraft Instrument Dial, Standard Form of |
| MS33586 | Metals, Definition of Dissimilar |

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be encountered in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.5 Corrosive fumes. The materials, as installed in the indicator and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.6 Protective treatment. When materials are used in the construction of the indicator that are subject to



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deterioration when exposed to climatic and environment conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The indicator shall be designed to display pitch and roll attitude information with full freedom about each axis and turn and slip information.

3.4.1 Electronic components. Electronic components and the application thereof shall be in accordance with the class 2 requirements of MIL-E-5400.

3.4.2 The indicator shall be so designed and constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases. The complete mechanism, consisting of all components except the pitch trim knob assembly, shall be removable from the case as a unit. The required lighting may or may not be considered a part of the complete mechanism, depending on the contractors's design. The indicator shall be built to withstand the strain, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.3 Reliability program. The contractor shall establish a reliability assurance program in accordance with MIL-R-26667.

3.4.3.1 Reliability. The indicator shall have a minimum mean-time-between-failures (MTBF) of not less than 750 hours at a 90 percent confidence level which shall be demonstrated during reliability testing.

3.4.3.2 Longevity. The indicator shall have a minimum operating life of 5,000 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF requirements.

3.4.4 Maintainability. Maintainability shall be in accordance with MIL-M-26512. The mean active maintenance downtime shall not exceed 20 hours. The confidence level shall be 90 percent and the accuracy level shal be 85 percent.

3.4.4.1 The following qualitative requirements shall be considered in the design of the indicator:

a. Minimization of complexity of maintenance tasks (for example: calibration, adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly, or component

d. Minimum number and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all components requiring maintenance, inspections, removal, or replacement.

3.5 Performance. The indicator shall meet the requirements specified herein under the following conditions:

a. Operating temperatures ranging from -54° to +71°C and storage temperatures down to -62°C

b. Relative humidity up to 100 percent



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c. Pressures ranging from 30 inches Hg down to 0.81 inch Hg (sea level to 80,000 feet)

d. Exposure to salt sea atmosphere

e. A potential of 500V dc for a period of 10 seconds

f. Vibration with a double amplitude of 0.10 inch from 5 to 14 cps and ±1g (peak) acceleration from 14 to 500 cps

g. Fungus growth as encountered in tropical climates

h. Sand and dust particles as encountered in desert areas

i. Rainfall as encountered in any locale

j. Acceleration forces up to 10g

k. Voltage and frequencies varying from 103 to 127V and 380 to 420 cps, respectively.

3.5.1 Power consumption. The primary power consumption shall not exceed 26 va; power consumption for white or red lighting shall not exceed 10 va.

3.5.2 Radio noise interference. The indicator shall meet the radio noise interference requirements of MIL-I-6181.

3.5.3 Sensitivity. The sensitivity of each axis shall be 1/10 of a degree or better.

3.5.4 Magnetic property. When held in various positions magnetically east or west of and not more than 12 inches from the center of a free magnet in a magnetic field having a horizontal intensity of 0.18 ±0.01 oersted, deflection of the magnet shall not exceed 5°.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

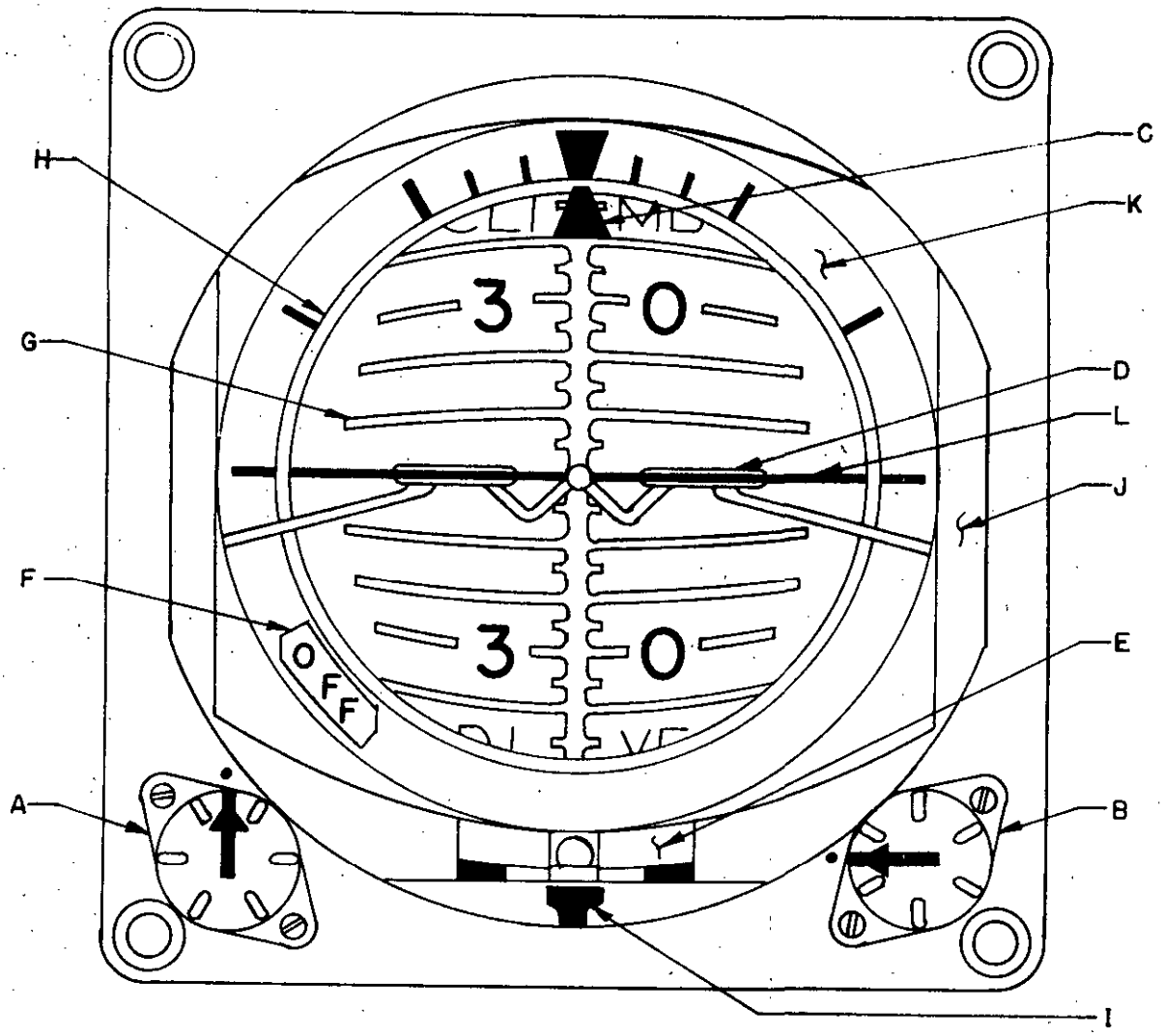
3.7 Component parts. The indicator shall consist of the parts and mechanisms necessary to provide and operate an attitude display, an inclinometer, and rate of turn display as specified herein and illustrated on figures 1, 2, and 3. All colors referenced shall be in accordance with FED-STD-595.

3.7.1 Display. The display shall basically consist of a bank-angle dial, bank indexes, rate-of-turn pointer, inclinometer, power failure indicator, miniature aircraft symbol, and an attitude sphere background which is supported and rotated about two servo-driven gimbal axes. All interior visible parts that are not part of the display shall be finished in durable dull black as approved by the procuring activity.

3.7.1.1 The bank scale dimensions and markings shall be in accordance with figures 1, 2, and 3. The front of the inner edge of the dial shall not exceed 3/4 inch from the inner surface of the cover glass. The portion of the dial within ±45° of the zero-bank index shall be a portion of a cone with the visible portion of the dial at approximately a 75° angle to the roll axis of the indication. The front edge of the outer periphery of the dial within ±45° of the zero-bank index shall not exceed 11/16 inch from the inner surface of the cover glass. Unless otherwise specified, the front edge of the outer periphery of the dial shall not exceed 1 inch from the inner surface of the cover glass. The outer radius of the dial shall be 1-29/32 inches.



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A. ROLL TRIM KNOB
B. PITCH TRIM KNOB
C. BANK ANGLE INDEX
D. MINIATURE AIRPLANE
E. INCLINOMETER
F. POWER FAILURE INDICATOR
G. SPHERE
H. MECHANISM MASK
I. RATE OF TURN
J. DIAL MASK
K. BANK ANGLE DIAL & SCALE
L. HORIZON BAR

FIGURE 1. Dial Display



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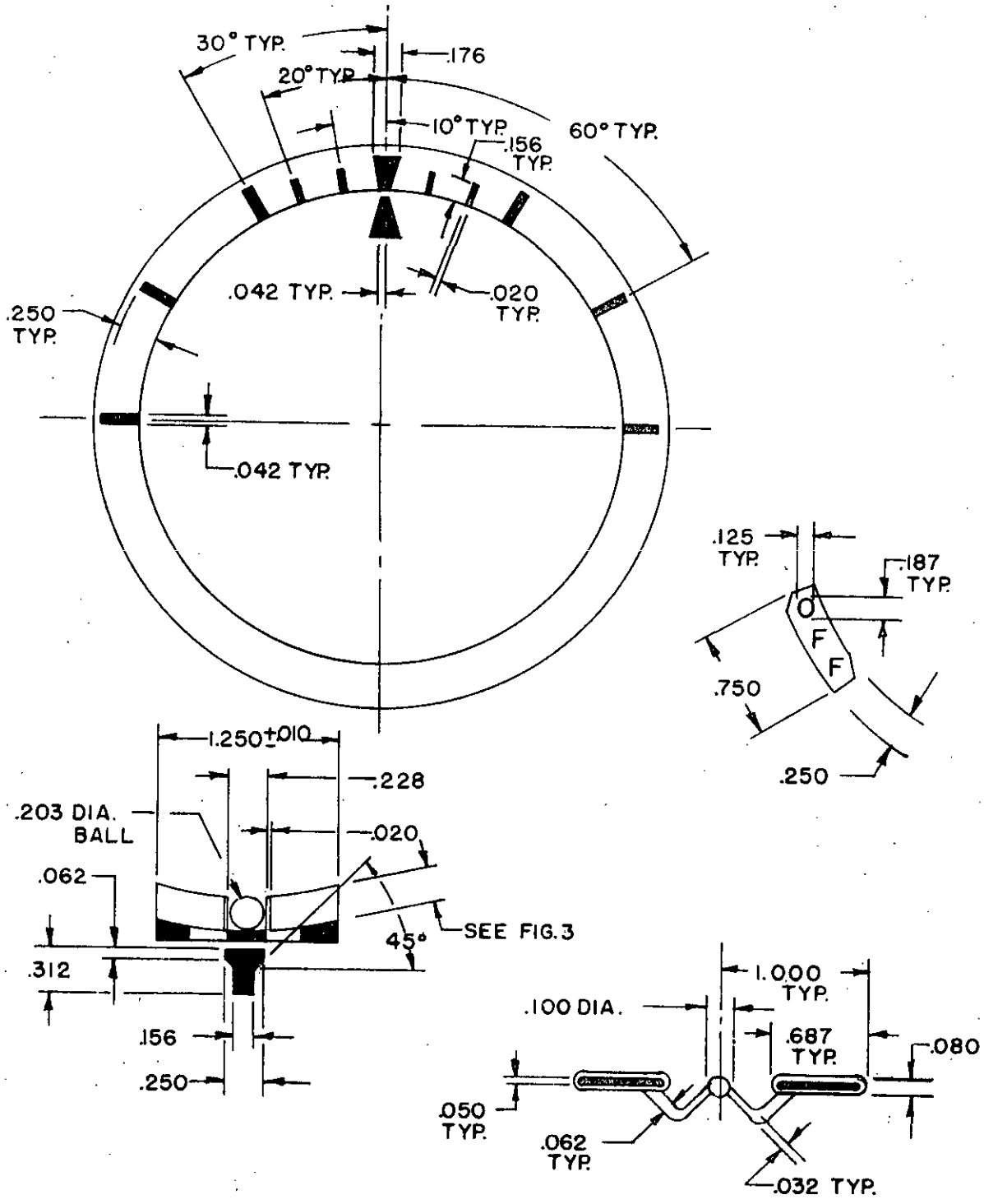



FIGURE 2. Details



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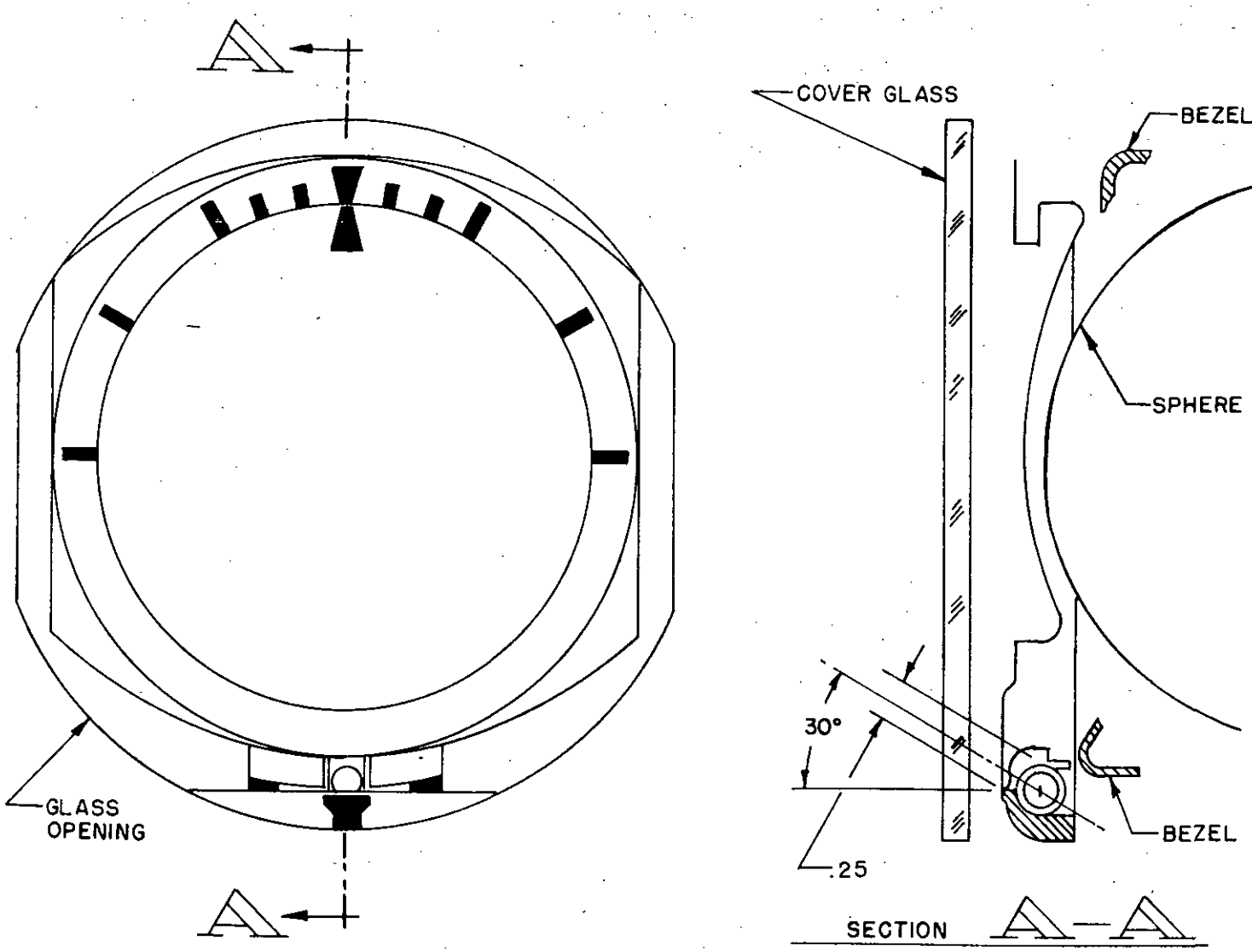



FIGURE 3. Roll Mask And Assembly



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3.7.1.1.1 Dial masks. A mask shall be attached to each side of the bank scale dial in accordance with figures 1, 2, and 3.

3.7.1.2 Mechanism mask. A mask shall be incorporated which is attached to the roll gimbal and extends essentially from the bank scale to the attitude sphere. The visible surface of this mask shall be shaped as the surface of a frustrum of a cone. The large diameter shall be forward and shall be approximately 3-9/32 inches. The edge of the mask shall be hidden by the inner diameter of the bank scale. The small diameter shall be toward the rear and shall be approximately 3-5/32 inches. There shall be approximately 1/16 inch between this edge of the mask and the sphere.

3.7.1.3 Bank index. The bank index shall be affixed to and shall rotate with the roll gimbal. The bank index shall be in accordance with figures 1 and 2. At zero roll indication, the index shall be adjacent to and aligned with the zero roll marking on the bank scale.

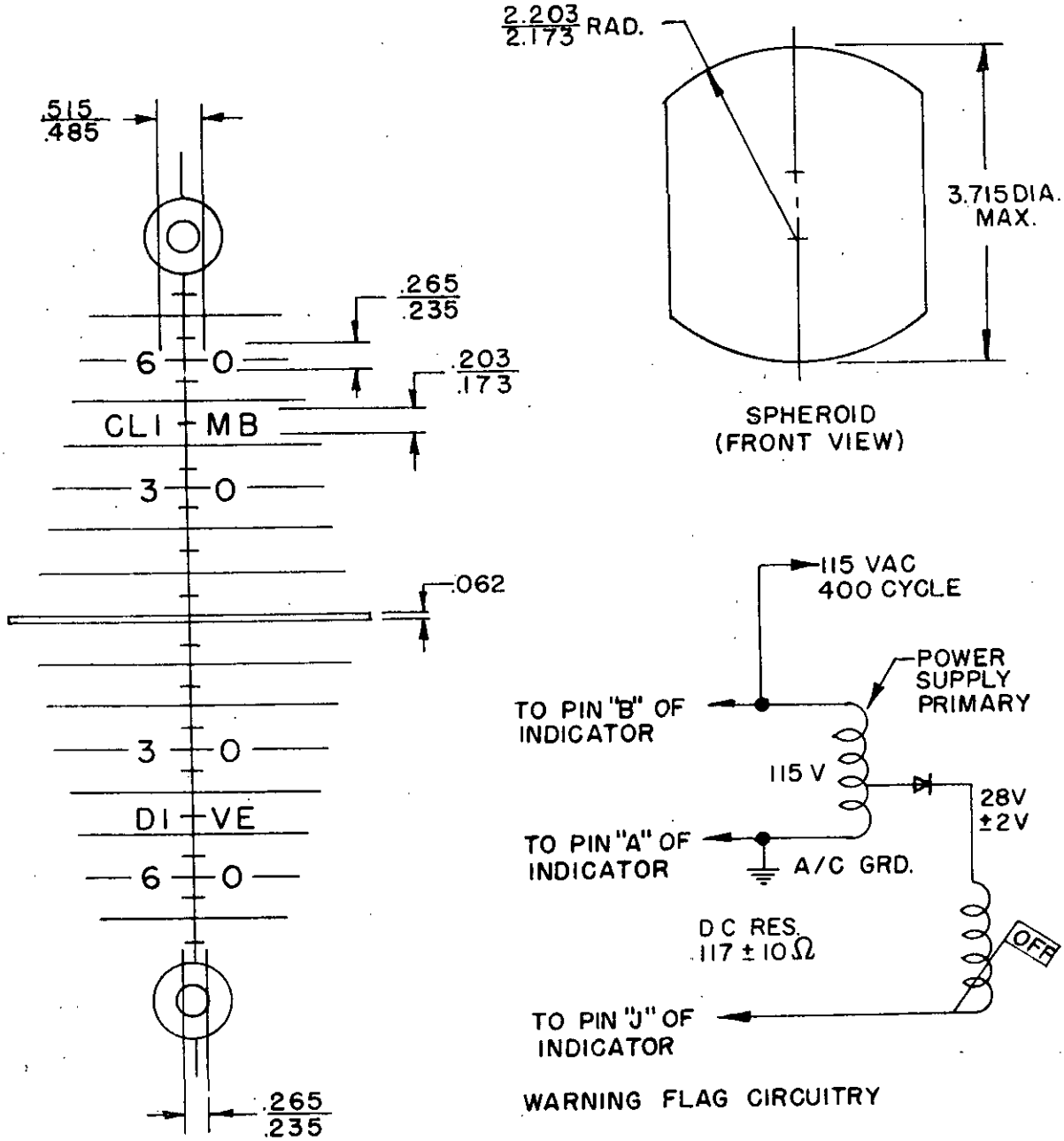
3.7.1.4 Miniature aircraft. The miniature aircraft shall be in accordance with figure 2. The indicator shall be so designed that the miniature aircraft will be located between the attitude sphere and the light wedge. The front surface of the center dot shall be not more than 0.060 inch maximum from the sphere surface. The center of this dot shall be 2-1/2 ± 1/32 inch from the top edge of the mounting flange. The wings of the miniature aircraft shall be curved so that they follow the curvature of the sphere.

3.7.1.5 Attitude sphere. The attitude sphere that forms the background of the display shall be mounted in the roll gimbal of the indicator. It shall rotate with the gimbal about the roll axis for roll indications and shall rotate in relation to the roll gimbal about the pitch axis for pitch indications. The roll axis shall be perpendicular to the face of the indicator and the pitch axis. The pitch axis shall be horizontal when the indicator is level and is indicating a zero-roll attitude. The sphere and sphere markings shall be in accordance with figure 4.The sphere shall be driven in pitch so that the horizon line goes above the miniature aircraft for dive maneuvers and below for climb maneuvers. The distance between the cover glass and the sphere shall be held to a minimum and shall not exceed 25/64 inch.

3.7.1.5.1 Pitch trim. Means shall be incorporated for adjusting the position of the sphere by means of a pitch trim knob mounted on the lower right-hand side of the face of the indicator and marked in accordance with figures 1 and 5. The knob shall have approximately one-fourth turn of freedom in the clockwise (cw) direction and one-half turn of freedom in the counterclockwise (ccw) direction from the zero pitch trim position. The freedom in each direction shall be limited by a positive stop. When the knob is rotated to the stop in the clockwise direction, the sphere shall rotate to deflect the horizon line upward to indicate a dive of between 5° and 10°. When the knob is rotated to the stop in the counterclockwise direction, the sphere shall rotate to deflect the horizon line downward to indicate a climb of between 10° and 20°. The servo system shall operate properly for any pitch trim adjustment. A minimum of 1 ounce-inch and a maximum of 8 ounce-inches shall be required to rotate the pitch trim knob through the full range of settable indications in both the cw and ccw directions. The torque required shall not exceed the above range in any of the environments or reliability tests in which the indicator is specified to operate properly.



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THE PITCH REFERENCE LINES SHALL BE SO THAT THEY ARE PORTIONS OF GREAT CIRCLES WHICH INTERSECT AT THE PITCH AXIS

FIGURE 4. Spheroid And Markings



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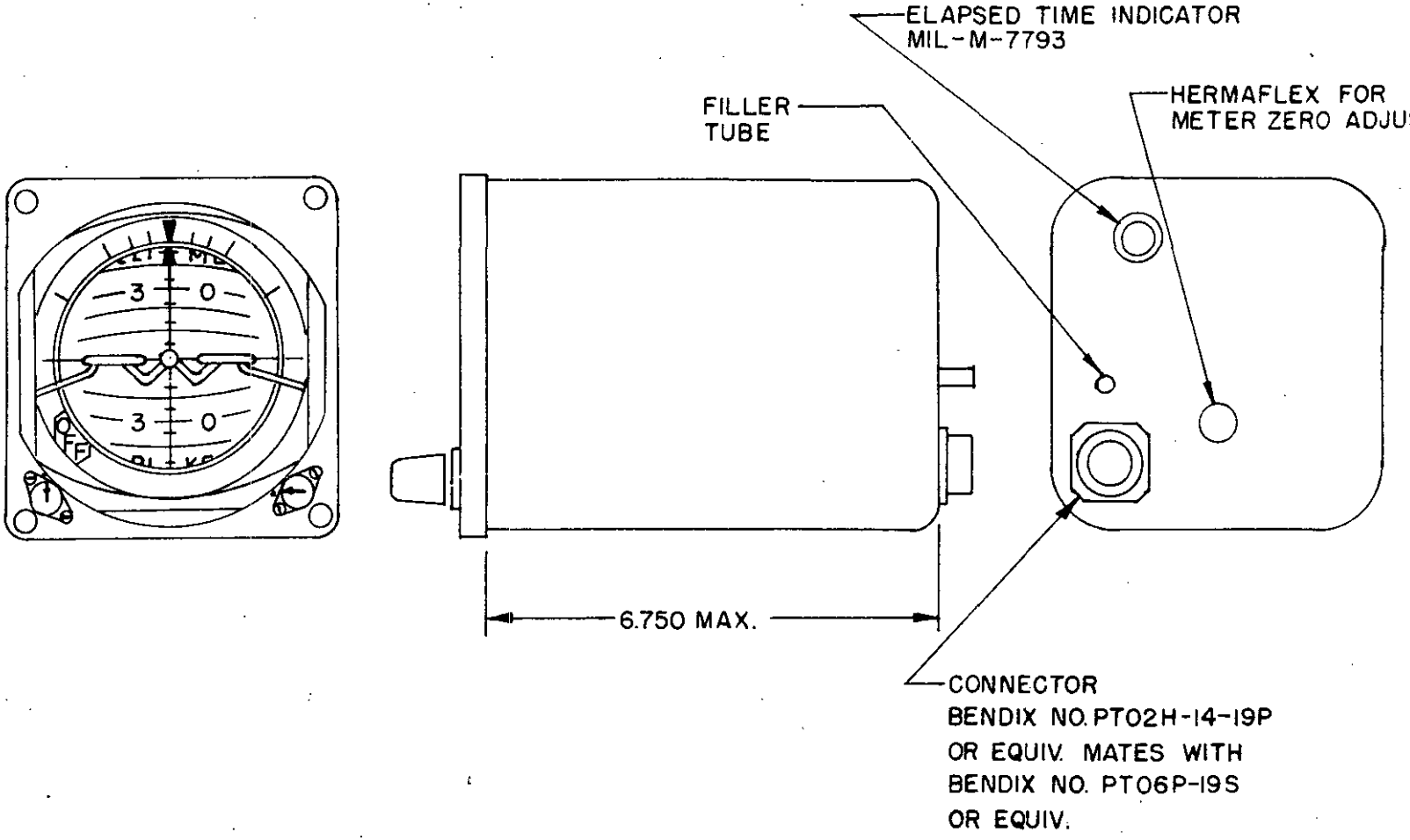



FIGURE 5. Case

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3.7.1.5.1.1 Roll trim adjustment. A roll trim adjustment shall be provided for adjusting the position of the bank index from 8° to 20° right and left bank. The roll trim adjustment shall be located on the front face of the instrument as shown on figure 1.

3.7.1.5.1.2 The design shall be such that when two indicators are hooked in parallel, the change in the trim adjustment of one indicator will not affect the indication on the second instrument. The design shall be such that no degradation of performance will result at any pitch trim or roll zero setting. The pitch and roll trim knobs, items A and B, figure 1, shall be medium gray, color No. 36231. The arrow provided on the knob shall be black, color No. 37038, and the reference mark against which the arrow on the knob is set shall be white, color No. 37875.

3.7.2 Indicator synchro control transformer. The synchro control devices on the indicator axes shall be high impedance Eclipse-Pioneer type AY500-5 synchros, or equal. The servo circuit shall be so designed that when one to four indicators per axis are connected to a single test transmitter synchro, no noticeable change in performance can be noted. All synchro control device stator leads shall be electrically isolated from each other, from other circuitry in the case, and from the case.

3.7.2.1 The standard test transmitter synchro shall be an Eclipse-Pioneer type AY201-1, or equal, with minimum and known errors. These errors shall be subtracted from those of the indicator under test.

3.7.2.1.1 The transmitter shall be equipped with an accurate dial having a ±1° vernier and readable to 0.1°. It shall be graduated through 360° and shall be capable of continuous rotation.

3.7.2.1.2 The transmitter shall be calibrated and the rotor and stator leads identified as follows:

a. The rotor leads designated as H and C with C grounded

b. One stator lead designated as Z and connected to C

c. 26V 400 cps applied across H and C and the voltage measured across the two remaining stator leads and across H and one of the stator leads

d. The rotor turned until the voltage across H and the one stator lead is approximately maximum (about 35V) and the voltage across the two stator leads is exactly minimum. The dial locked to the rotor with the dial exactly on zero

e. The dial turned through increasing numbers and the voltage measured across the two undesignated stator leads and Z. The voltage across XZ will increase before decreasing and the voltage across YZ will decrease before increasing.

3.7.2.1.3 Test transmitter excitation. The test transmitter rotors shall be excited from the same power supply voltage that operates the indicator. It shall be so applied that the voltage from rotor lead (C) to rotor lead (H) will be in phase with the voltage from pin A to pin B of the indicator with lead (C) and pin A (see table I) common.

3.7.2.2 Indication zeroing and sensing. When the pitch and roll servo control devices of the indicator are connected to standard test transmitters in accordance with table I with the respective transmitters set at the 180° dial reading, the indications shall come to rest at zero. Increasing dial readings of the respective test transmitter shall produce right-roll and pitchup indications. Decreasing dial readings shall produce opposite indications.



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TABLE I Power Input Connections

| Connector Pins | Function |
|---|---|
| A | Neutral (external power ground) |
| B | 115V 400 cps single-phase power in phase with excitation to test transmitter synchros |
| C | Spare |
| D (X), E (Y), F (Z) | Roll servo control transformer stator leads |
| R (X), H (Y), P (Z) | Pitch servo control transformer stator leads |
| J | Power failure warning (connect externally to pin A to complete circuit) |
| K | Internal lighting power input for red lighting (0-5V) |
| L | Neutral for internal lighting |
| T | Internal lighting power input for white lighting (0-5V) |
| U (+), V (-) | Rate of turn indicator (deflects to right) |
| M | Pitch null test (pitch error voltage between this pin pin and pin A) |
| N | Roll null test |
| G | No connection |
| S | No connection |

3.7.2.3 Attitude actuation. The sphere shall be rotated about the pitch and roll gimbal axes by means of geared servo motors until a null is attained between the synchro control devices on the indicator axes and the transmitter synchros in the pitch and roll displacement reference. The freedom of sphere rotation about either axis shall not be limited.

3.7.2.4 Followup rate. The rate at which the motors drive the gimbals shall be not less than 300° per second in roll nor less than 90° per second in pitch.

3.7.2.5 Accuracy. The pitch and roll indications shall agree with the input signals from their respective test



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transmitter synchros to within ±1/2° for all positions through 360°.

3.7.2.6 Followup operation. The electrical and mechanical design of the indicator shall be such that in the process of followup, the indications of pitch and bank will move smoothly without hunting, oscillating, or jumping more than 1/10° total amplitude in pitch or bank when (1) each servo input is connected to standard transmitters for individual testing of each, and (2) when each servo input receives a standard synchro input which simulates all possible pitch and bank motions and rate of motions of an aircraft. Interaction between the two axes will not be permitted. There shall be no sustained electrical or mechanical oscillations in either axis at any attitude nor overshoot greater than 3° with a step input applied equal to the maximum followup lag.

3.7.2.7 Amplifiers. The amplifiers provided as a part of the indicator shall be 3-stage transistorized amplifiers with the second and third stages redundant transistor stages. The amplifiers shall be designed with redundant circuits so that the indicator will continue to operate, though with reduced performance, should partial failure of the amplifier occur.

3.7.2.8 Zeroing adjustments. Means shall be incorporated to externally adjust the position of the sphere ±3° minimum in pitch and roll. The initial setting of the pitch and roll indication at the time of delivery of the indicator shall indicate zero pitch and roll within ±1/2° when the indicator is connected to a test transmitter set to the 180° dial reading. The pitch adjustment shall be combined with the pitch trim circuit by means of a settable knob. The trim adjustment shall be equipped with stops so positioned that an erroneous indication of attitude cannot result from the positioning of this adjustment. The trim adjustment shall incorporate a locknut or similar device to secure the adjustment setting.

3.7.3 Power failure indicator. A power failure warning indicator shall be incorporated in the lower left of the face of the indicator as shown on figures 1 and 2. The word OFF shall be black on a red background. A lead shall be brought out through the electrical connector as shown in table I. When the ground connection or the power is removed, or both, the warning shall appear. For night lighting, the flag shall be front lighted.

3.7.3.1 Compatibility. The warning flag shall operate satisfactorily when connected in parallel with one to three indicators having the warning flag circuitry shown on figure 4.

3.7.4 Electrical power requirements. The power supplied to the indicator shall be 115V, 1 phase, 400 cps in accordance with MIL-STD-704.

3.7.4.1 Electrical power variation. The indicator shall operate properly and continuously at any a-c voltage between 103 and 127V, any frequency between 380 and 420 cps, or any combination thereof, in addition to transient variations as specified in MIL-STD-704.

3.7.4.2 Case isolation. The case shall be isolated from all electrical circuits within it.



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3.7.5 Inclinometer. The inclinometer shall be incorporated into the display in accordance with figures 1, 2, and 3.

3.7.5.1 Tube. The glass of the inclinometer tube shall be made of clear annealed glass tubing free from any flaws that will seriously affect the readability of the inclinometer. The inside of the tube shall be smooth and uniform so that the ball may roll freely except for the damping of the liquid. The arc length of the visible portion of the tube shall be approximately 1-1/4 inches. The radius of curvature of the tube shall be 3-1/2 inches.

3.7.5.2 Ball. The ball shall be a highly polished, accurate, 13/64-inch white sphere. A minimum of two-thirds of the ball shall remain in view at either extreme of the tube when viewed from 12 inches in front of the zero mark of the slip indicator at an angle of 30° with the horizontal plane prescribed by the base of the indicator.

3.7.5.3 Damping liquid. The damping liquid used shall be sufficiently colorless to preclude interference with the visibility of the ball under daylight and night lighting conditions and under normal and temperature extremes.

3.7.6 Rate of turn. The rate of turn display shall be in accordance with figures 1, 2, and 3. The moving index shall be mounted as a meter movement that is pivoted at a point approximately 4-1/2 inches behind the pointer. The design shall be such that the pointer will be as stable as practicable under rough-air conditions. Within normal operating position, the index shall move left or right in agreement with the polarity of the direct current received from the remote rate of turn sensor. An external mechanical zero adjustment shall be provided. The scale shall be made to conform to the path of the index so that there is a minimum of parallax. The front surface of the scale and index shall be within 1/4 inch of the cover glass at zero displacement. The index deflection to either side of the center zero position shall be limited to 5/8 inch. At each of the fully deflected positions, a minimum of one-half of the index shall be in view when viewed at a horizontal distance of 2 feet from the center dot of the display.

3.7.6.1 Deflection (rate of turn index). The direction of deflection for a given input shall be as specified in table I.

3.7.6.2 Response (rate of turn index). The index response shall be linear in degrees with respect to current to within 7.5 percent of the proportionate full-scale value. A deflection of one index width from the zero position shall require 0.5 ma ±7.5 percent. A deflection of two index widths so that the index aligns with the appropriate outer scale mark shall require 1.0 ma ±7.5 percent.

3.7.6.3 Resistance (rate of turn index). The resistance across the rate-of turn input terminals shall be 1,000 ohms ±3 percent at 25°C.

3.7.6.4 Damping (rate of turn index). The pointer mechanism shall be so damped that there will be not more than 1.5 percent overshoot.

3.7.6.5 Response time (rate of turn index). Response time (see 3.7.6.2) of the rate-of-turn index shall be a maximun of 1/3 second.

3.7.7 Elapsed time indicator. An MS17322-2A or -6A elapsed time indicator, or equal, shall be incorporated to record the running time of the indicator proper. The elapsed time indication shall be provided in such a manner that it will not be necessary to unseal the indicator proper to read the elapsed time recorded.



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3.8 Markings. Unless otherwise specified, the dimensions of all markings on the indicator shall be in accordance with figures 1, 2, 3, and 4. Unless otherwise specified, all visible portions of the indicator shall be color No. 37038 in accordance with FED-STD-595.

3.8.1 Numerals and letters. Unless otherwise specified, all numerals and letters on the indicator, other than the nameplate, shall be in accordance with MS33558.

3.9 Electrical connectors. A connector, Scintilla Division, Bendix Aviation Corporation type PT02H-14-19P, or equal, shall be provided on the indicator in the manner shown on figure 5. The connections shall be in accordance with table I. This connector shall mate with a Scintilla Division connector type PT06A-14-19S, or equal.

3.10 Cover glass and lighting wedge. The cover glass or lighting wedge shall be flat and free from discolorations, scratches, and striae that might interfere with reading of the indicator. The glass shall be tempered, shall be mounted in the flange by insertion within the case, and shall be a minimum of 3/16 inch thick. All reflecting glass surfaces shall be provided with a reflection-reducing coating that meets the requirements of MIL-C-675, in addition to withstanding the environmental conditions specified herein, except that the following reflectance tolerances shall apply:

| Angle of Incidence | Wavelength-Millimicrons | Percent reflectance |
|---|---|---|
| 0° | 450 - 675 | 0.6 absolute |
| 0° | 425 - 700 | 0.5 average |
| 30° | 450 - 625 | 1.0 absolute |
| 30° | 425 - 700 | 0.5 average |

3.11 Indicator case. The indicator case shall be in accordance with MS33545, with the exceptions stated herein. Dimension A of MS33545 shall be in accordance with figure 1. Dimension L of MS33545 shall be the length of the indicator as shown on figure 5. The pitch trim knob shall be approximately 3/4 inch in diameter and 1 inch long. The case of the indicator proper shall be hermetically sealed. The filling tube shall not exceed 0.34 inch in length and shall be located as shown on figure 5.

3.11.1 Filling medium. When a hermetically sealed case is used, the filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 100 percent helium, a mixture of 88 to 92 percent nitrogen and the remainder helium, or 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practicable, the 100-percent-helium filling medium shall be utilized.

3.11.2 Leak rate. The indicator shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential of approximately 1 atmosphere.

3.12 Lighting. Internal lighting shall be designed into the indicator. The design of the indicator shall be such that the internal lighting may be either white or red lighting, depending upon the pins shown in table I to which the power for lighting is applied. With the exceptions specified herein, the red lighting shall be in accordance with MIL-L-25467 and the white lighting shall be in accordance with MIL-L-27160. The ratio of the brightness of the gray portion of



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the sphere to the brightness of the aircraft shall never exceed 0.6. The internal lighting of the prime standard and all production indicators shall be subject to the approval of the procuring activity. All production units shall be visually compared to the prime standard for approval.

3.12.1 Lamps. MS25237-328, MS24515-718-AS15, or MS24367-715-AS15 lamps shall be used as the light source.

3.12.2 Prime standard. The internal lighting shall be comparable to and shall balance in brightness with a prime standard indicator. The prime standard shall be either an attitude-director conforming to the requirements of this specification, or a lighted model. The prime standard shall be equipped with both red and white lighting and shall be used as the reference for acceptance of all production indicators.

3.12.2.1 A minimum of 12 average areas covering all representative areas of the face of the prime standard indicator shall be measured for brightness in foot-lamberts. The prime standard instrument shall be measured periodically (at least every 50 hours of operation) to determine any brightness change. Any variation greater than ±20 percent in two or more of these areas will require the return of the prime standard to the procuring activity for investigation, reapproval or corrective action, or both.

3.12.2.1.1 Visual comparison. The general overall lighting level of the indicator shall fall within the limits of the prime standard when energized at the voltage levels specified (see 4.6.21.1).

3.12.3 Stray light. The lighting system shall be so housed as to prevent the leakage of stray light and to shield all lamp filaments from direct view. Stray-light measurements shall be made as follows: A flat white reflecting surface of 85 ±5 percent reflectance shall be placed perpendicular to the front cover glass and parallel to the top edge of the mounting flange. Brightness readings shall be made looking down (perpendicular) on the reflecting surface and 1.0 inch in front of the cover glass. When the reflecting surface is positioned in the lower half of the display, the readings shall not exceed 1.5 foot-lamberts and when positioned in the upper half, shall not exceed 0.2 foot-lambert. Measurements shall be repeated looking up on the reflecting surface. In no case shall the readings exceed 0.2 foot-lambert.

3.13 Soldering. Soldering shall be accomplished in accordance with MIL-S-6872.

3.14 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.15 Lubrication. The indicator shall be lubricated in accordance with MIL-STD-838.

3.16 Weight. The weight of the indicator shall not exceed 6.0 pounds.

3.17 Finishes and protective coatings

3.17.1 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.17.1.1 Dials, small holes, and case inserts need not be anodized.

3.17.1.2 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.



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3.17.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.17.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.17.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.17.1.6 Parts in hermetically sealed containers need not be anodized.

3.17.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.17.2 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.18 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 Workmanship. The indicator, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assembly screws and bolts.

3.19.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.2 Cleaning. The indicator shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.19.3 Dimensions. Dimensions and tolerances not specified shall be close as is consistant with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.19.4 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt threads.

3.19.5 All markings shall be clear and sharp. Dimensions shall be as specified and on any given mark the width shall not vary noticeably throughout the length of the mark. Markings which for some given indication become adjacent and are specified to have the same basic dimension at the adjacent point shall not be noticeably different at that point.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.2 Classification of tests. The inspection and testing of the indicators shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Unless otherwise specified, tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C).

4.3.2 Voltage input. Unless otherwise specified, the indicator shall be tested with 115V 400-cps (nominal) single-phase power.

4.3.3 Proper connection. When the indicator is referred to as being properly connected, it shall mean that it is connected to a test fixture providing a lighting circuit voltage variable from 5V ac or dc to approximately 0V, synchro signals in accordance with 3.7.2 through 3.7.2.2, variable-current d-c signals 1,000-ohm source and with reversible polarity to the meter movements, a-c power as required, and that pin j is jumped to pin A through a closed switch.

4.3.4 Visual observation. Unless otherwise specified, all visual observations of indicator performance shall be made at a distance of 2 feet within a sighting cone 30° from the normal to the surface of the cover glass.

4.3.5 Recordings. Upon application of power, the indication of the elapsed time indicator shall be recorded. Readings shall be recorded after completion of the tests specified in 4.6.28, 4.6.30, 4.6.31, 4.6.35.1, 4.6.35.2, and 4.6.36 and after completion of all tests specified under 4.6.

4.3.6 Operating position. Unless otherwise specified, the indicator shall be tested in the normal operating position.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of 6 indicators representative of the production equipment. Three samples shall be subjected to all the tests except reliability and the other three samples shall be subjected to the reliability test only. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 Preproduction tests. The preproduction tests shall consist of all of the tests specified under 4.6.

4.5 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Burn-in

b. Examination of product

c. Starting

d. Pitch and bank zero

e. Sensitivity

f. Followup rate



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g. Followup accuracy

h. Followup operation

i. Indication lag

j. Hunting and jumping

k. Overshooting

l. Gimbal freedom

m. Pitch trim

n. Roll trim

o. Zeroing adjustments

p. Sealing

q. Power-failure indicator

r. Inclinometer sensitivity

s. Inclinometer zero position

t. Inclinometer friction

u. Inclinometer visibility

v. Inclinometer filling

w. Response

x. Lighting

y. Fogging.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Three indicators shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests (with the exception of burn-in)

b. Power consumption

c. Low voltage

d. Magnetic property

e. Acceleration

f. Dielectric

g. Vibration error

h. Horizon line and bank index null position shift

i. Low temperature operation

j. High temperature operation.

k. Slip indicator leak

l. Inclinometer damping, room temperature

m. Response time

n. Antireflection test.

4.5.2.2 Sampling plan B. Unless otherwise specified, 3 indicators selected at random from the first 15 items of the contract or order shall be subjected to the following tests in the order specified and as described under 4.6:

a. Sampling plan A

b. Radio noise interference

c. High temperature exposure

d. Temperature-altitude (cycling)

e. Temperature shock

f. Humidity

g. Fungus

h. Rain

i. Salt fog

j. Sand and dust



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k. Vibration failure

l. Acceleration

m. Sealing thermal shock

n. Structural

o. Pitch trim torque

p. Pitch trim effect

q. Pressurization.

4.5.2.2.1 At the option of the procuring activity, the sampling plan B tests may be divided into two groups as follows, to be conducted on two sets of three instruments each if a need exists to expedite testing. Each indicator shall meet the individual tests after completing the group of specific tests to which submitted.

| Group I | Group II |
|---|---|
| Radio noise interference | Humidity |
| High temperature exposure | Fungus |
| Temperature-altitude cycling | Rain |
| Temperature shock | Salt fog |
| Pitch trim effect | Sand and dust |
| Vibration failure | Sealing thermal shock |
| Acceleration | |
| Structural test | |
| Pitch trim torque | |
| Pressurization | |

4.5.2.3 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of indicators on hand or later produced shall not be made until it is determined that the indicators meet all the requirements of the specification.

4.5.2.4 Sampling plan C. Unless otherwise specified, 3 indicators shall be selected at random from the first 15 produced on the contract or order and subjected to the preproduction reliability test specified in 4.6.37.1. Acceptance or rejection shall be determined as specified therein.

4.5.2.5 Sampling plan D. Unless otherwise specified, 3 indicators shall be selected at random from the first 20 indicators of each production quantity of 100 that have already passed the individual tests and subjected to the acceptance reliability test specified in 4.6.37.2. Acceptance or rejection shall be determined as specified therein.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking. The indicator shall be placed in the face-down position, shaken by hand, and inspected in this position to determine that no loose hardware or foreign material is present. Evidence of any such material shall be cause for rejection.



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4.6.2 Starting. The indicator shall be properly connected. When power is applied, the warning flag shall disappear and the indicator shall be ready for immediate operation.

4.6.3 Pitch and bank zero. The standard test transmitters in the pitch and roll axes shall be set to the 180° dial reading. Power shall be applied and the indicator shall follow up to the zero indications in pitch and bank within ±1/2° from any random indication in either axis. This test shall be repeated for any initial indication at the discretion of the procuring activity. The pitch and roll indications shall be adjusted to zero by means of the zeroing adjustments specified in 3.7.2.8.

4.6.4 Sensitivity. The test transmitter for the roll axis shall be moved 1/4° in right bank and returned to the original position and then 1/4° in left bank and returned to the original position. This test shall be repeated in climb and dive with the pitch trim adjusted to its extreme positions except that the pitch test transmitter shall be moved 1/2° in each direction. The test transmitter for the pitch axis shall be moved 1/4° in dive and returned to the original position and then 1/4° in climb and returned to the original position. There shall be a smooth, perceptible movement of the indicator bank index and horizon line for each movement of the appropriate test transmitter.

4.6.5 Followup rate. The indicator shall be properly connected to a remote signal source or standard test transmitter. The followup rates in pitch and roll shall be determined. The rate in the roll axis shall be not less than 300° per second. The rate in the pitch axis shall be not less than 90° per second with the pitch trim at zero and when adjusted to each of its extreme positions.

4.6.6 Followup accuracy. The standard test transmitter in the roll axis shall be displaced to at least 10 random positions other than the 180° dial reading. The indicator shall position itself with the rotor of the standard test transmitter within ±1/2° for each position within ±30° indication and within 1° at all other indications. The standard test transmitter in the pitch axis shall be displaced to at least 10 random positions other than the 180° dial reading. The indicator shall position itself with the standard test transmitter within ±1/2° for each position within ±30° indication and within ±1° for all other indications.

4.6.7 Followup operation. Suitable test equipment shall be connected to the appropriate test pins of the indicator and any hunting, jumping, lag, or overshooting of the indication when the indicator is made to follow the standard transmitter at various constant rates in both the pitch and roll axes shall be observed and recorded. This shall be possible with the indicator sealed. The data shall be recorded during the initial individual tests preceding all other tests and during the final individual tests following the completion of all other tests. While performing the individual tests at other specified times, the indicator may be checked visually for noticeable hunting, jumping, lag or overshooting.

4.6.7.1 Indication lag. The indication shall not lag behind the shaft of the remote input transmitting synchro by more than the amount shown in the curve on figure 6. The indicator shall be tested at three different rates (including the maximum rate specified in 4.6.5) for each servo. The servos may be checked at any rate within the maximum rate, at the discretion of the procuring activity. A 25 percent increase in indication lag shall be permitted during temperature extremes and low voltage tests.

4.6.7.2 Hunting and jumping. There shall be no noticeable hunting or jumping



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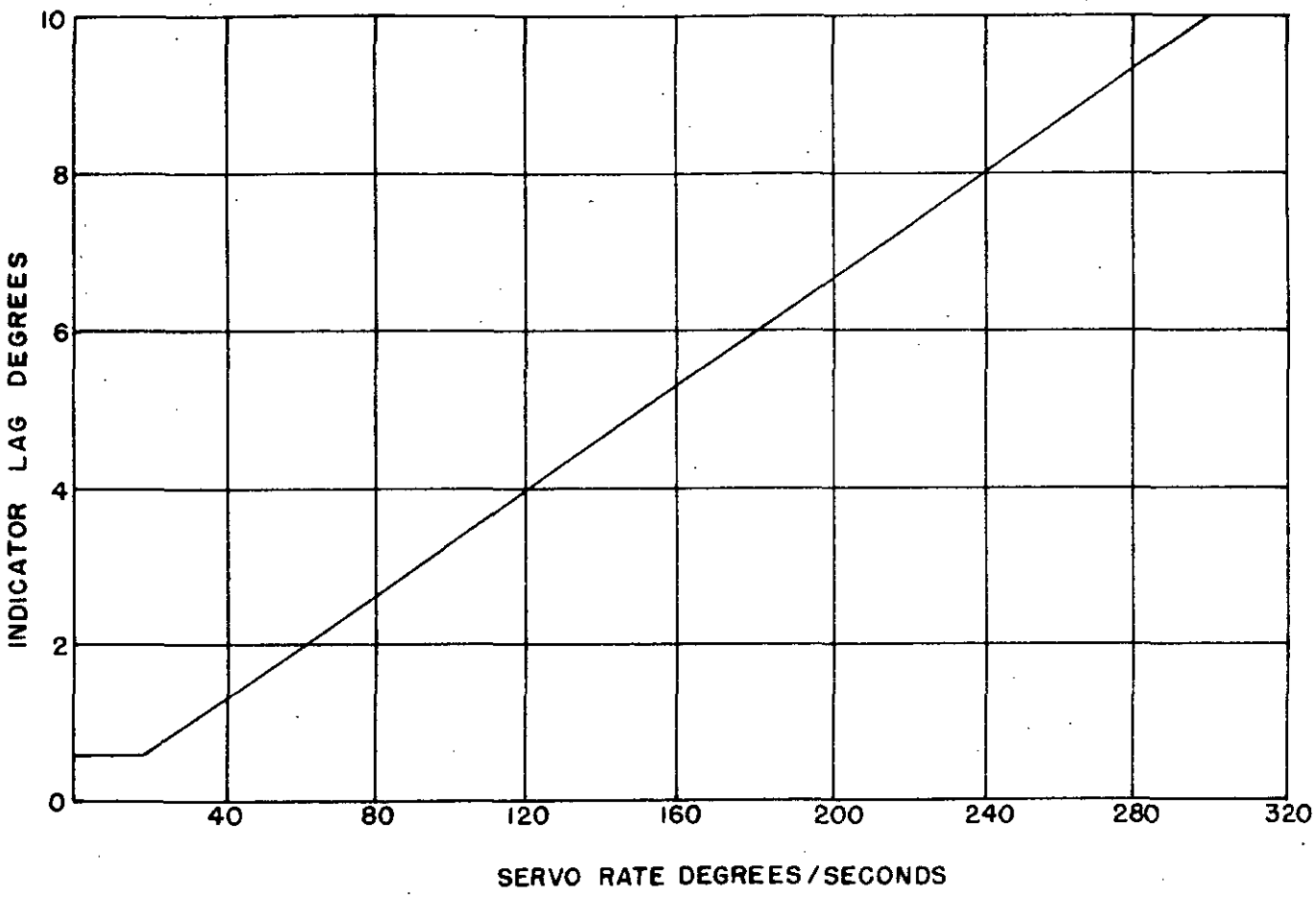



FIGURE 6. Servo Rate Vs Indicator Lag

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when the standard transmitter shaft is not rotating. The manufacturer shall conduct a survey on the indicator of his design operated over the full range of operating speeds. The testing rates chosen for hunting and jumping shall be those that produce the most erratic motion and shall include at least 4 rates within each of the following ranges; 0° to 20° per second, 20° to 90° per second, and 90° to 300° per second. The peak-to-peak amplitude of the hunting and jumping shall not exceed 1/10° for any shaft speed up to 90° per second in the pitch axis or 300° per second in the roll axis. The maximum shaft speed at which the test is conducted in the pitch axis shall be 90° per second and the maximum shaft speed at which the test is conducted in the roll axis shall be 300° per second. The rates selected shall be submitted to the procuring activity for approval. Any hunting, jumping, except as specified above, or interaction noted during this test or any other test specified herein shall be considered failure.

4.6.7.3 Overshooting. Upon completion of pitch or roll maneuvers, overshooting shall be held to a practical minimum and shall not exceed one overswing with a maximum amplitude of 3° with a step input equal to the maximum allowable followup lag.

4.6.8 Gimbal freedom. The indicator shall be properly connected and the remote signal source or standard test transmitter moved 360° in roll and pitch. The indicator shall follow smoothly without sticking and overshooting or hunting greater in magnitude than that specified in 4.6.7. There shall be no disturbance of the meter movements as a result of this test.

4.6.9 Pitch trim. The pitch trim knob shall be adjusted to its limits and the deflection of the horizon line observed. When the knob is turned in a cw direction, the indicator shall show a 5° to 10° dive and when turned in a ccw direction, the indicator shall show a 10° to 20° climb.

4.6.10 Roll trim. The indicator shall be properly connected with power applied. When the test signal is at electrical zero, the bank index shall indicate zero within ±1/2°. When the knob is rotated in a cw direction to the stop position, the indicator shall show 8° to 20° right bank and when rotated in a ccw direction to the stop position, the indicator shall show 8° to 20° left bank.

4.6.11 Zeroing adjustments. The indicator shall be properly connected to the standard test transmitters with each set to electrical zero. The roll zeroing adjustment shall be moved to its extremes and deflection of the bank index shall be not less than ±3° from the zero position. The pitch test transmitter shall be rotated 3° in climb and then 3° in dive. In each case, the pitch trim knob shall be loosened on its shaft and the shaft rotated until the indication is zeroed. The pitch trim knob shall be tightened again to the shaft at the zero position. In each case, the test specified in 4.6.9 shall be met. This test shall be performed only during the initial individual tests and during the final individual tests conducted as part of the reliability test.

4.6.12 Sealing. The case of the indicator proper shall be tested for leaks by means of a mass spectrometer type of helium leak detector. The maximum leak rate shall not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential of approximately 1 atmosphere.

4.6.13 Power failure indicator. The indicator shall be properly connected but no power shall be applied. Power



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shall then be applied and the flag shall disappear. The power shall be disconnected and the flag shall immediately appear. Power shall be reapplied and the jumper from pin j to pin A shall be disconnected. The flag shall appear.

4.6.13.1 Compatibility. The indicator shall be connected in parallel with three other attitude indicators having a warning flag circuitry conforming to figure 4. When the lead which parallels all J pins to indicators is not connected to the aircraft ground, or pin A, the warning flag shall appear. Conversely, when the lead that parallels the J pins is connected to pin A, the warning flag shall disappear.

4.6.14 Inclinometer sensitivity. With the dial vertical, the indicator shall be rotated about the roll axis to the right until the ball is just short of its limit. The angle of rotation shall be 10° ±2°. The test shall be repeated rotating the instrument to the left. The same tolerance shall apply. The ball shall not stick at the high end of the tube.

4.6.15 Inclinometer zero position. The indicator shall be positioned with the upper edge of the mounting bezel horizontal and vibrated at 0.001 inch double amplitude at 10 cps. The inclinometer ball shall rest at the zero mark of the slip indicator within ±1/32 inch.

4.6.16 Inclinometer friction. The inclinometer ball shall roll smoothly when the indicator is tipped to either side in the vertical plane of the dial at the rate of 2° $^{+0.1°}_{-0.0°}$ per second throughout its entire range.

4.6.17 Inclinometer visibility. The indicator shall be so tipped that the ball is at rest at either end of the tube. Not less than two-thirds of the ball shall be visible when the ball is viewed from the position specified in 3.7.5.2.

4.6.18 Inclinometer filling. The indicator shall be so tipped that all the air in the tube is trapped in the expansion chamber end of the tube. The face of the indicator shall be in a vertical plane with the line joining the centers of the two lower mounting holes in a horizontal plane. No air bubbles shall be visible when the indicator is viewed from the position specified in 3.7.5.2.

4.6.19 Response. The response of the rate-of-turn pointers shall be measured and shall be in accordance with 3.7.6.2.

4.6.19.1 Response time. The response time of the indicating pointer shall be measured and shall be in accordance with 3.7.6.5. For this test, the instrument shall be shunted externally with a resistance of 350 ohms to allow for the shunting effect of another indicator in parallel with the usual source. The voltage source shall be steady and shall be applied to the above circuit through at least 50,000 ohms. At least 5 seconds shall be allowed in order to establish the calibration of the 100 percent mark.

4.6.20 Pointer balance. With the indicator in a normal upright position and the pointer mechanism de-energized, the rate-of-turn pointer shall not deflect more than 1/64 inch as the indicator is rotated 360° in roll and pitch.

4.6.20.1 The indicator shall be accelerated with a force of 10g along its normal vertical axis. The rate-of-turn pointer shall not vary by more than 5/64 inch from the zero position. This test shall be repeated with the acceleration applied along each of the two axes that are perpendicular to the vertical axis and to each other. The rate-of-turn pointer position shall not vary more than 5/64 inch from the zero position.

4.6.21 Lighting. The lighting circuits of the indicator shall be properly connected and the indicator tested in accordance with MIL-L-25467 or



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MIL-L-27160, as applicable. Upon completion of the reliability test and other environmental tests as specified, slight degradation in the brightness and color readings shall not be considered as failures unless so determined by the procuring activity. Lighting data (brightness and color) shall be taken at the start of the reliability test and at least every 500 hours thereafter. In addition, the indicator shall meet the requirements specified in 3.12.

4.6.21.1 Visual comparison. The prime standard and the indicator under test shall be placed side by side at eye level. The two indicators shall be viewed at a distance of approximately 2 feet in a sighting cone 30° from normal to the surface of the cover glass. The overall lighting of the two units shall be compared with 3.00 ±0.01V applied to the lighting terminals of the indicator under test. General overall lighting level comparison shall then be made between the production indicator being tested and prime standard with the prime standard voltage at 2.85V and then at 3.15V. All production units failing the visual comparison tests shall have the questionable areas measured with a photometer as specified in the applicable lighting specifications MIL-L-25467 and MIL-L-27160. If they also fail this test, the production indicators shall be rejected. Visual comparison by prime standard shall not be used for testing of preproduction units. If the general overall lighting level of the indicator does not fall within the brightness limits of the prime standard when energized at the specified voltage levels, the indicator shall be rejected.

4.6.22 Fogging. The indicator shall be energized for a minimum period of 1 hour in a 71° ±2° C controlled ambient. After this period and while at this temperature, ar ice cube shall be rubbed on the indicator glass face for a period of from 1 to 2 minutes. The glass shall be wiped dry (do not use compressed air) and the indicator inspected for evidence of water or oil fog. Evidence of fogging shall be cause for rejection.

4.6.23 Power consumption. The power consumption of the indicator shall be measured with suitable meters. The a-c consumption shall not exceed 26 va. The power factor shall be maintained between 0.70 and 1.00. The power consumption and power factor shall be measured with the pitch servo operating at 90° per second and the roll servo operating at 300° per second.

4.6.24 Low voltage. The indicator shall be properly connected. The a-c voltage shall be reduced to 103V and 320 cps. The indicator shall meet all individual tests, except 4.6.11, 4.6.14, 4.6.15, 4.6.16, 4.6.17, 4.6.18, 4.6.19, 4.6.20, and 4.6.21. This test shall be repeated at all combinations of 380 to 420 cps and 103 to 127V.

4.6.25 Magnetic property. This test shall be made first with no power applied to the indicator and then repeated with the indicator operating on rated power. The indicator shall be held in various positions magnetically east or west of and not more than 12 inches from the center of a free magnet approximately 1-1/2 inches long in a magnetic field with a horizontal intensity of 0.18 ±0.01 oersted. The maximum deflection of the magnet shall not exceed 5°. An aircraft compass with the compensating magnets removed may be used as the free magnet for this test.

4.6.26 Acceleration tests. The indicator shall be properly connected and power applied to the indicator. The indicator shall be accelerated along its normal vertical axis with a force of 10g. The attitude display shall not vary by more than 1° in pitch or roll from the initial indication and the pointer shall not vary by more than 5/64 inch



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from the zero position. This test shall be conducted with the attitude set at zero pitch and roll and with each axis displaced 90°.

4.6.27 Dielectric. With the indicator properly connected, a potential of 500V dc shall be applied between isolated pins of Scintilla connector PTO2H-14-39P, or equal and between pins of this connector and the indicator case for a period of 10 seconds. The resistance shall be not less than 10 megohms.

4.6.28 Vibration error. The indicator shall be properly connected and subjected to a vibration error test in accordance with MIL-STD-810, method 514.1, figure 514-1. The vibration levels and frequencies shall be as follows:

| Frequency | Level |
|---|---|
| 5-14 cps | 0.10 double amplitude |
| 14-500 cps | ±1g (peak) acceleration. |

The frequency sweep shall be made between 5 and 500 cps. During this test, the indicator shall meet the following:

4.6.28.1 Roll, pitch, and pointer. Oscillation of the attitude sphere in both pitch and roll throughout the frequency range shall not exceed 1/2° in roll and pitch and oscillation of the rate-of-turn pointer shall not exceed one-half the width of the pointer.

4.6.28.2 Roll, pitch and pointer variation. Throughout the frequency range, oscillation of the attitude sphere in pitch and roll during vibration shall not vary from the indication before vibration by more than 1/2° and oscillation of the rate-of-turn pointer shall not vary more than 1/2 the width of the pointer.

4.6.28.3 Oscillation and variation tests may be conducted at any indication at the discretion of the procuring activity.

4.6.28.4 Inclinometer accuracy. The slip indicator reading shall not differ at any time by more than 1/16 inch from the reading obtained when the indicator is at rest.

4.6.29 Horizon line and bank index null position shift. The indicator shall be properly connected to standard test transmitters in the pitch and roll axes. The transmitters shall be set to drive the indicators in both axes to the zero position. The input voltage and frequency shall be varied within the following limits and combinations thereof:

| | |
|---|---|
| Frequency | 380 to 420 cps |
| A-C Voltage | 103 to 127V |

The horizon line shall not shift more than ±1/16 inch from normal null position. The bank index shall not shift more than 1/4° from normal null position.

4.6.30 Low temperature operation. With no power applied, the indicator shall be properly connected and placed in a chamber at a temperature of -54° ±2° C for a 4-hour period. At the end of this period and with the indicator at the reduced temperature, power shall be applied for a period of 10 minutes after which the indicator shall meet the individual tests (except for burn-in and fogging tests) and the test specified in 4.6.32 except as follows:

a. The time of roll shall be 4 seconds maximum

b. The off-flag meter movement shall be checked only for operation without evidence of sticking or binding.

c. The tests specified in 4.6.11, 4.6.19, 4.6.20.1, and 4.6.21 need not be conducted until the indicator has returned to room temperature.



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4.6.31 High temperature operation. The indicator shall be properly connected and placed in a chamber at a temperature of 71° ±2°C and operated with power applied for a 4-hour period. At the end of this period and while at the high temperature, the indicator shall be subjected to and shall meet the individual tests (except burn-in test), with the following exceptions:

a. The off-flag meter movement shall be checked only for operation without sticking or binding

b. The tests specified in 4.6.12, 4.6.19, 4.6.20.1, and 4.6.21 need not be conducted until the indicator has returned to room temperature

c. The requirement for 1-hour exposure to 71° ±2°C prior to conducting the fogging test shall be considered to have been satisfied by the 4-hour exposure to 71° ±2°C.

4.6.31.1 Slip indicator leakage. Prior to the high-temperature test, the size of the air bubble in the tube shall be determined with the indicator at room temperature. Following the high-temperature test, the size of the air bubble shall again be determined with the indicator at room temperature. There shall be no appreciable change in the size of the air bubble.

4.6.32 Inclinometer damping, room temperature. The plane of the face of the indicator shall be placed in a vertical position with the upper flange inclined to the horizontal at an angle of 12°. The indicator shall be tipped suddenly through a vertical angle of 24° so that the ball will roll to the opposite end of the tube. The time of roll of the ball from the zero mark of the indicator to the end of the tube shall be between 0.15 and 0.50 second.

4.6.33 Burn-in. The following tests shall be conducted once on every indicator furnished under this specification. These tests shall be conducted after the indicator has been calibrated and hermetically sealed and prior to the other individual tests conducted in accordance with 4.5.1. During these tests, no adjustment shall be made. Failure to pass any of these tests shall be cause for rejection. The tests of 4.6.33.1 and 4.6.33.2 may be conducted separately or concurrently.

4.6.33.1 Servo loops. The pitch and roll servo loops shall be excited and subjected to an input which shall drive both axes through a minimum of ±30° from 5 to 7 times per minute. This test shall be conducted continuously for 20 hours. The indicator shall be examined every 4 hours for evidence of hunting, jumping, or interaction.

4.6.33.2 Rate-of-turn pointer. A current of 1,000 micro-amperes dc shall be applied as a step input to pins P (+) and R (-) followed by a step input of the same magnitude to pins P (-) and R (+). This current step input shall be repeated in each polarity for a total of 10 cycles. No step input shall start until the rate-of-turn pointer has returned to the zero position. Upon completion of 10 cycles in each direction, the rate-of-turn pointer shall be tested as specified in 4.6.18.1 to determine compliance with 3.7.6.1, 3.7.6.2, and 3.7.6.4.

4.6.34 Radio noise interference. The radio noise interference test shall be conducted in accordance with MIL-I-6181. Any out-of-specification condition shall be investigated and necessary action taken before retesting is begun.

4.6.35 Environmental tests. The indicator shall be subjected to the following tests, conducted in accordance with MIL-STD-810 and as specified.



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4.6.35.1 High temperature exposure. The indicator shall be properly connected and subjected to the high temperature exposure test in accordance with method 501.1 for a 48-hour period with no power applied. At the end of this period and while still at the high temperature, the indicator shall meet the test specified in 4.6.8. The off-flag meter movement shall operate without binding or sticking. At the conclusion of this test with the indicator at room temperature, the indicator shall be chekced for satisfactory operation. Satisfactory operation is defined as proper operation of all servos and the off-flag meter without sticking or binding at any indication.

4.6.35.2 Temperature-altitude (cycling). The indicator shall be properly connected and tested in accordance with method 504.1, except that step 7 shall be conducted at 71°C and step 10 shall not be conducted. Satisfactory operation is defined as proper operation of all servos and the off-flag meter movement without sticking or binding at any indication. At the conclusion of this test and after the indicator has returned to room temperature, it shall meet all individual tests except burn-in.

4.6.35.3 Temperature shock. The indicator shall be subjected to a temperature shock test in accordance with method 503.1 after which it shall meet all individual tests except burn-in.

4.6.35.4 Humidity. The humidity test shall be conducted in accordance with method 507.1. At the conclusion of this test, the indicator shall meet the tests specified in 4.6.7 and 4.6.19. There shall be no evidence of corrosion or rust of internal or external parts as a result of this test.

4.6.35.5 Fungus. The indicator shall be subjected to a fungus test in accordance with method 508.1 after which the indicator shall meet the tests specified in 4.6.7 and 4.6.20. There shall be no evidence of deterioration that might adversely affect subsequent operation nor shall fungus be supported.

4.6.35.6 Rain. The indicator shall be subjected to a rain test in accordance with method 506.1 except that the indicator shall be mounted in a rain test chamber for 1/2 hour with the dial in a vertical plane and for 1/2 hour with the dial in a horizontal plane (dial face upward). Upon removal from the chamber, the indicator shall meet the tests specified in 4.6.7 and 4.6.19.

4.6.35.7 Salt fog. The salt fog test shall be conducted in accordance with method 509.1 for a period of 50 hours. At the end of this period, the indicator shall meet the tests specified in 4.6.7 and 4.6.19. There shall be no damage as a result of this test and the tolerances specified for the individual tests shall apply.

4.6.35.8 Sand and dust. The sand and dust test shall be conducted in accordance with method 510.1. At the end of this test, the indicator shall be removed from the test chamber and allowed to cool at room temperature. The indicator shall then meet the tests specified in 4.6.7 and 4.6.19.

4.6.35.9 Vibration failure. The indicator and amplifier shall be subjected to a vibration test in accordance with method 514.1, table 514-I, 1C1A. The amplifier shall be connected to the indicator by cabling and tested in accordance with method 514.1, table 514-I, 1A1C, at room temperature. No looseness in the mechanism nor damage to any part of the indicator shall result from this test. At the conclusion of this test, the indicator shall pass all individual tests except the burn-in test.



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4.6.36 Acceleration. The indicator (not operating) shall be mounted in its normal operating position and accelerated first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The indicator shall be subjected to an acceleration of 20g in each of the above axes for a period of 1 minute. At the end of this acceleration, the indicator shall meet all individual tests except burn-in. No damage to the indicator shall result from this test.

4.6.37 Reliability

4.6.37.1 Preproduction. Three indicators shall be subjected to a reliability demonstration test in accordance with MIL-R-26667 during which a 750-hour MTBF at a 90 percent confidence level shall be demonstrated. The three indicators shall each be operated for 2,000 hours for a total test time of 6,000 hours. If 6 or more failures occur prior to the accumulation of the 6,000 hours of total test time, the equipment shall be rejected. The test environment shall be in accordance with MIL-R-26667, table entitled "Environmental Test Levels", test level VI, except that the chamber temperature shall be -35°C in lieu of -54°C and vibration shall be ±1g in lieu of ±2g. Reporting shall also be in accordance with MIL-R-26667. During each operational period and with each indicator properly connected, the following conditions shall exist:

a. Both axes of the attitude display shall be driven ±30° from 0° pitch and roll indications respectively. During the first 1,000 hours of operation on each indicator, the pitch axis shall receive a 10° per second input while the roll axis shall receive a 12° per second input. Throughout the final 1,000 hours of operation on each indicator, the pitch axis shall receive a 1° per second input and the roll axis shall receive a 2° per second input during odd numbered operating periods and 10° per second in pitch and 12° per second in roll respectively during even numbered operating periods. Within 10 minutes after the beginning of each operating period, the sphere shall follow the input signals without showing evidence of hunting, jumping, or interaction throughout the entire ±30° range or at the reversal points.

b. The lighting circuits shall be energized with 5V ac or dc. The red lights shall be operated during the first 1,000 hours of testing on each indicator and the white lights shall be operated during the final 1,000 hours of testing on each indicator.

c. The power failure indicator shall disappear immediately after the beginning of each operating period.

d. The rate-of-turn pointer shall be driven through its complete operating range at a rate of 1/2 inch per second. Within 10 minutes after the beginning of each operating period, the pointer shall move smoothly over its entire range.

e. Once each day, the indicators shall be observed for the first 15 minutes of a given operating period and once each hour thereafter during that operating period to determine if any of the indicator's functions are not operating properly. At least two of these observations shall be made during vibration periods.

4.6.37.1.1 During the cooling-off period between each operating period, (1) all power, including power to the rate-of-turn pointer, lighting circuits, and flag circuits, shall be removed from the indicator under test, (2) the standard transmitters shall remain excited and their rotors shall be displaced through a maximum of ±30° at a rate of 10° per second, and (3) the pitch trim knob shall be rotated to its extremes in both directions 20 times.



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4.6.37.1.2 After every 200 ±10 hours of operating time on each indicator and at the conclusion of the reliability testing, each indicator shall pass all individual tests except the burn-in test.

4.6.37.2 Longevity. At the completion of the testing specified in 4.6.37.1 through 4.6.37.1.2, testing of one of the three indicators shall be continued for an additional 3,000 hours on each in accordance with MIL-R-26667. Each failure shall be recorded and the equipment repaired prior to resuming this test.

4.6.37.3 Acceptance. An acceptance reliability test shall be conducted to demonstrate that the production indicators have a 750-hour MTBF at a 90 percent confidence level. The three indicators shall each be operated for 2,000 hours for a total test time of 6,000 hours. The test environment and conditions shall be as specified in 4.6.37.1. If 2 or fewer failures have occurred after 3,000 hours of the total 6,000-hour test time has elapsed, the equipment may be accepted on the condition that testing will be continued to the 6,000-hour limit and the contractor will assume total responsibility for delivered equipment if the 750-hour MTBF is not demonstrated. If 3 or more failures occur prior to the accumulation of 3,000 hours of the total 6,000-hour test time, or 6 or more failures during the 6,000-hour test time, the contractor shall modify the equipment to eliminate the cause of failure and begin a new acceptance test using modified indicators. The contractor may incorporate corrective action during any portion of the test provided such action is first submitted to the procuring activity for analysis and approval. Subsequent to corrective action, all equipment produced on the contract shall be modified by the contractor to the approved configuration.

4.6.38 Sealing thermal shock. The indicator shall be immersed alternately in tap water maintained at 85° ±5°C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes. Not more than 5 seconds shall elapse between bath immersions. No damage to the hermetic seal shall result due to this test. Following the immersions, the indicator proper shall be tested for leaks by means of a mass-spectrometer-type of helium leak detector. The initial maximum detected leak rate at a differential pressure of 1 atmosphere shall not permit more than 10 percent contamination of the filling medium after 1,000 hours.

4.6.39 Structural test. Neither the indicator electrical connector nor the indicator case shall be damaged or deformed in any way when a total force up to 60 pounds is applied to the connector along the longitudinal axis of the instrument while inserting the indicator into a stationary mount upon installing the indicator in the aircraft. The testing apparatus and procedure to be utilized shall be approved by the procuring activity prior to conducting this test.

4.6.40 Pitch trim torque. The torque required to rotate the pitch trim knob shall be measured throughout the range of settable indications in both the cw and ccw direction and shall be from 1 to 8 ounce-inches. A minimum of 5 readings shall be taken.

4.6.41 Pitch trim effect. Two indicators shall be connected in parallel to the same power source with both pitch synchro control transformers connected to the same standard test transmitter. The standard test transmitter shall be an Eclipse-Pioneer type AY200-1, or equal, and shall be connected as specified in 3.7.2.1 and 3.7.2.2 except that a calibration dial need not be provided. The pitch transmitter stator leads shall



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also be connected to a third synchro control transformer, Eclipse-Pioneer type AY500-5, or equal, having a 10,000-ohm resistive load connected across the rotor leads. The following test procedure shall apply:

a. With pitch trim knob of each indicator set to a visual indication of zero pitch trim, the test transmitter rotor shall be positioned to produce a 0° (±5°) pitch indication on the two indicators.

b. The rotor of the third synchro control transformer shall then be positioned to produce null voltage across the control transformer rotor winding of 100 millivolts, or less, as measured by a vacuum tube voltmeter.

c. Rotation of the pitch trim knob of either indicator from the zero trim position to the extreme position in either direction shall not produce a change in the null voltage across the third control transformer rotor of more than ±20 millivolts from the initial null voltage reading. The pitch trim knob of the other indicator shall be maintained in the zero position during this test.

d. This test shall be repeated with the test transmitter rotated 180° ±5° from the original position.

4.6.41.1 All the tests of 4.6.41 shall be conducted as outlined using an Eclipse-Pioneer type AY200-3, or equivalent, as the test transmitter.

4.6.41.2 All the tests of 4.6.41 shall be conducted as outlined using a Kearfott CM01012-107, or equivalent, as the test transmitter.

4.6.42 Pressurization. The indicator shall be placed in a chamber at an absolute pressure of 28 pounds per square inch and a temperature of 25°C for a period of 4 hours. During this period, no deformation or other damage to the indicator shall occur. At the conclusion of this test, the indicator shall operate satisfactorily. Evidence of deformation, damage, or any unsatisfactory operation shall be cause for rejection. After this test, the indicator shall pass the leak rate test.

4.6.43 Reflectance. The reflectance of all reflecting glass surfaces shall be measured and the reflectance tolerances shall be within those specified in 3.10.

4.6.44 Maintainability verification. Maintainability verification shall be in accordance with Appendix A to MIL-M-26512.

## 5. PREPARATION FOR DELIVERY

### 5.1 Preservation and packaging

5.1.1 Level A. When the electrical connector is protected by a waterproof-vaporproof cap with gasket, the indicator shall be preserved in accordance with MIL-P-116, method III. When the waterproof-vaporproof cap is not utilized, method 1A-14 of MIL-P-116 shall be used. The indicator shall be cushioned on all sides, top, and bottom with 5 inches of bound fiber conforming to MIL-C-7769, type III, or equal.

### 5.2 Packing

5.2.1 Level A. Indicators preserved and packaged to meet 5.1.1 shall be packed in exterior type shipping containers conforming to PPP-B-636, type I or type II, class 2.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows:

Indicator, Attitude ARU-18/A, Director



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6. NOTES

6.1 Intended use. The ARU-18/A attitude director indicator covered by this specification is intended for use in helicopters. The indicator will be controlled by a vertical reference control that provides transmitting synchro pitch and roll output signals and a rate gyro providing signals for the d-c meter movement.

6.2 Ordering data. Procuring documents should specify the following:

a. Title, number, and date of this specification

b. Written report of failures occurring during testing to be submitted to the procuring activity within 48 hours after failure

c. When sampling plan B, C, and D tests will not be conducted.

6.3 Definition

6.3.1 Hermetic seal. Hermetic sealing is the process by which an item is totally enclosed by a suitable metal structure or case and sealed airtight by fusion of metallic or ceramic materials. This includes the fusion of metals by welding, brazing or soldering; the fusion of ceramic materials under heat or pressure; and the fusion of ceramic materials into a metallic support. Elastomeric or resinous materials or combinations of these materials may be used as a cover glass seal provided that the specific leak-rate requirements are met under all tests and environmental conditions listed herein. Elastomeric or resinous materials may be used in other areas provided prior approval is obtained from the procuring activity. These materials shall be in accordance with MIL-S-7502, type B-2, or MIL-A-8623, type II or III. If these materials are used, the surfaces to be bonded must be cleaned in accordance with MIL-A-9067. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

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